

THREATLENS – PROJECT DETAILS

1. Project Domain

Cybersecurity / Artificial Intelligence and Machine Learning

ThreatLens is a cybersecurity-based project that uses Artificial Intelligence and Machine Learning techniques to identify and analyze potential security threats associated with personal information.

2. Problem Statement

Personal information such as email addresses, phone numbers, and online identifiers can be targeted by attackers for phishing, spam, fraud, identity theft, and other cyber threats. Many users do not have an easy way to understand whether their information may be exposed to potential security risks.

ThreatLens aims to develop a system that analyzes user-provided information, identifies potential cybersecurity threats or suspicious patterns, and presents the analysis in a simple and understandable manner.

3. Project Objectives

Objective 1

To analyze personal information and identify potential security risks or suspicious patterns.

Objective 2

To apply Machine Learning and Natural Language Processing techniques for detecting and classifying possible cybersecurity threats.

Objective 3

To provide a simple and user-friendly interface that displays the threat analysis and security-related information in an understandable format.

4. Dataset Name and Source

Dataset Name

Phishing and Threat Intelligence Dataset

Possible Sources

- **PhishTank** – phishing URL data
- **OpenPhish** – phishing and malicious URL information

- **Public phishing email/URL datasets** available through research repositories

The dataset can contain features such as URLs, domain information, text/email content, suspicious keywords, and other characteristics useful for identifying phishing or malicious activity.

Note: The final dataset should match the actual dataset used in the ThreatLens implementation.

5. Algorithms

The following Machine Learning algorithms can be used for ThreatLens:

1. **Logistic Regression** – used for binary classification such as safe or suspicious.
 2. **Random Forest** – an ensemble learning algorithm useful for classification and handling multiple features.
 3. **Naive Bayes** – particularly useful for text and email classification.
 4. **Support Vector Machine (SVM)** – useful for classification of high-dimensional data.
 5. **TF-IDF** – can be used as an NLP feature-extraction technique for analyzing text.
 6. **CNN/LSTM** – can be considered if deep-learning-based text or sequence analysis is included.
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6. Tools and Technologies

Programming Language

- **Python**

Libraries and Frameworks

- Pandas
- NumPy
- Scikit-learn
- Natural Language Processing (NLP)
- Flask / FastAPI

Frontend

- HTML
- CSS
- JavaScript

Development and Version Control

- Git

- GitHub
 - Jupyter Notebook / Google Colab
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7. Expected Outcome

The expected outcome of ThreatLens is a **user-friendly cybersecurity threat analysis system** that can:

- Analyze user-provided information.
- Detect suspicious patterns.
- Identify potential phishing or security risks.
- Classify information as potentially safe or suspicious based on the implemented model.
- Present the analysis in an easy-to-understand format.
- Increase user awareness about possible cybersecurity threats.

The system is intended to act as an **awareness and threat-analysis tool**, rather than guaranteeing that information is completely safe or unsafe.

8. Relevant Research Papers

Research Paper 1

Title: *Machine Learning Based Phishing Detection from URLs*

Authors: Şahingöz et al.

Journal: Expert Systems with Applications, 2019

Relevance: This research studies machine-learning techniques for detecting phishing URLs and uses URL-related features for classification.

DOI: <https://doi.org/10.1016/J.ESWA.2018.09.029>

Research Paper 2

Title: *PhishNot: A Cloud-Based Machine-Learning Approach to Phishing URL Detection*

Authors: Alani & Tawfik

Journal: Computer Networks, 2022

Relevance: The research focuses on machine-learning-based phishing URL detection and evaluates classification approaches including Random Forest.

Source: <https://www.sciencedirect.com/science/article/pii/S1389128622004418>

Research Paper 3

Title: *Applications of Deep Learning for Phishing Detection: A Systematic Literature Review*

Year: 2022

Relevance: This paper reviews the application of deep-learning techniques such as DNN, CNN and RNN/LSTM for phishing detection.

Source: <https://pubmed.ncbi.nlm.nih.gov/35645443/>

Research Paper 4

Title: *Phishing URL Detection Using Machine Learning and Deep Learning*

Conference: IEEE World AI IoT Congress, 2024

Relevance: The research compares machine-learning and deep-learning approaches for detecting phishing URLs.

DOI: <https://doi.org/10.1109/AIIOT61789.2024.10579005>

Research Paper 5

Title: *Phishing URL Detection with Neural Networks: An Empirical Study*

Journal: Scientific Reports, 2024

Relevance: This study evaluates neural-network approaches for phishing URL detection and uses datasets including PhishTank and OpenPhish.